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Editorial Office
Computer Methods in Applied Mechanics and Engineering
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Dear Editor,

We submit the manuscript entitled “**PDE-Agents: An LLM-Orchestrated Multi-Agent Framework for Automated Finite Element Simulations with Knowledge Graph-Augmented Reasoning**” for consideration as an original research article in *Computer Methods in Applied Mechanics and Engineering*.

Computational-methods contribution. The manuscript introduces a new computational framework—not merely an application of existing tools—that changes how finite element simulations can be set up, executed, analysed, and iteratively improved. Specifically, we contribute:

1. A **multi-agent orchestration architecture** (LangGraph supervisor pattern) in which three specialist large language model (LLM) agents (Simulation, Analytics, Database) autonomously conduct the full FEM simulation lifecycle from natural-language task descriptions. This is the first system to combine LLM-based multi-agent orchestration with a production-grade FEM solver (DOLFINx/FEniCSx).
2. A **GraphRAG knowledge-augmented reasoning** layer (Neo4j knowledge graph with 768-dimensional vector embeddings) that provides agents with curated material properties, known failure patterns, boundary-condition statistics, and prior run lineage *before* committing to expensive compute. We introduce a novel *KG Smart* integration strategy combining warm-start injection with lazy conditional retrieval, and show that *integration pattern*—not knowledge content—determines whether GraphRAG augmentation helps or hinders an LLM agent.
3. A **rigorous evaluation framework** comprising: (i) formal V&V confirming $\mathcal{O}(h^2)$ spatial convergence across all benchmark cases; (ii) a three-way ablation study across 50 benchmark tasks demonstrating that KG Smart achieves 100% success rate with the highest output quality; (iii) a novel-material experiment proving KG value for genuinely unseen data (MPF = 1.00 vs. 0.34 for the KG-free baseline); and (iv) production-scale metrics from 1,369 real simulation runs showing 97.8% overall success.

Why CMAME. This work sits at the intersection of computational mechanics and AI-augmented scientific computing—a topic area explicitly within CMAME’s scope, which in-

cludes physically based machine-learning methods and digital-twin technologies. Our contribution is methodological: we propose, implement, and rigorously evaluate a new computational framework for automating FEM simulations, rather than merely applying an existing solver to a new problem. The benchmark suite includes cases where the method can plausibly fail (novel materials, ambiguous specifications, numerically challenging configurations), and we report explicit failure-mode analysis. We believe this work will be of broad interest to the CMAME readership working on computational automation, knowledge-augmented simulation, and AI for scientific computing.

Novelty and relation to prior work. While recent works such as ChemCrow and Co-scientist have demonstrated LLM agents for chemistry, no prior work combines multi-agent LLM orchestration with production FEM solvers and graph-based retrieval-augmented generation for computational mechanics. Our controlled ablation isolating integration pattern as the determining factor for GraphRAG utility is, to our knowledge, the first such result in the scientific computing literature.

Reproducibility. All code, models, evaluation scripts, and the knowledge graph schema are released openly at <https://github.com/MatPro-IFE/pde-agents>. The system runs entirely on locally deployed open-source LLMs (Qwen3-Coder-Next, Llama 4 Scout) without requiring proprietary API access.

Declarations. This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the manuscript and agree with its submission to CMAME. There are no conflicts of interest to declare.

Sincerely,

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